Lab: Profiling Your Inventive Agent

Objective

Map your own generative model as an inventor — identifying your prior beliefs, preferred states, identity priors, and blind spots. Then explore how collaborating with a different agent (a real or imagined partner) expands your inventive capacity. By the end of this lab, you will have a concrete "Inventor Agent Profile" that reveals your strengths, tendencies, and growth opportunities.

Materials and Prerequisites

- Your invention idea from Lab 01 (or a new one)
- A partner for Part 4 (can be a classmate, friend, colleague, or family member — ideally from a different background)
- If no partner is available, you will construct a hypothetical partner profile
- No technical background required

Part 1: Your Generative Model Inventory (15 minutes)

Every inventor carries a generative model — a set of beliefs about how the world works that determines what problems they notice and what solutions they imagine. Let us make yours explicit.

Step 1a: Domain Knowledge Map

What are the 3-5 areas where you have the deepest knowledge or experience? (These can be professional, hobbyist, or life-experience-based. Examples: cooking, carpentry, software, childcare, gardening, music, retail, nursing, teaching, etc.)

Write your response here

For each area, what cause-and-effect relationships do you understand well? (Example: "In baking, I understand how hydration ratio affects crumb texture" or "In retail, I understand how display placement affects purchase rates")

Domain	Causal Relationships I Understand

Write your response here

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Write your response here

Write your response here

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Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

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Write your response here

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Step 1b: Prior Beliefs Audit

List 5 assumptions you hold about how inventions work. These are beliefs you rarely question. Examples: "Good inventions must be simple," "Technology always involves electronics," "Users will read instructions," "The best solution is the cheapest one."

#	Prior Belief	How confident are you? (1-10)	Where did this belief come from?
1			

Write your response here

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Write your response here

|

Write your response here

| | 2 |

Write your response here

|

Write your response here

|

Write your response here

| | 3 |

Write your response here

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Write your response here

|

Write your response here

| | 4 |

Write your response here

|

Write your response here

|

Write your response here

| | 5 |

Write your response here

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Write your response here

|

Write your response here

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Step 1c: Solution Tendencies

When you imagine solutions to problems, what kind of solutions do you tend to generate? Check all that apply and add your own:

- [ ] Physical/mechanical devices
- [ ] Software/digital tools
- [ ] Organizational processes or systems
- [ ] Social interventions or community programs
- [ ] Chemical or material solutions
- [ ] Educational or instructional approaches
- [ ] Artistic or aesthetic improvements

- [ ] Economic or business model changes
- [ ] Other:

Write your response here

Why do you tend toward these types of solutions? What in your background explains this tendency?

Write your response here

Part 2: Your Preferred States — What Are You Trying to Achieve? (10 minutes)

Active Inference agents act to bring the world closer to their preferred states. Let us identify yours at multiple scales.

Scale 1: Immediate (this prototype, this week)

What would a successful next step for your invention look like?

Write your response here

Scale 2: Project (the completed invention)

Describe the world as it would be if your invention fully succeeds. What is different? Who benefits? How?

Write your response here

Scale 3: Identity (your inventive career/life)

What kind of inventor do you want to be? What values drive your inventive work?

Write your response here

Alignment Check

Are your three scales aligned? Does your immediate next step serve your project goal? Does your project goal serve your identity vision? If there are misalignments, describe them:

Write your response here

Part 3: Identifying Your Blind Spots (10 minutes)

The most dangerous part of a generative model is what it does not include — the variables, relationships, and possibilities that the agent cannot even perceive.

List 3 domains or skill areas you know very little about:

#	Domain I Know Little About	How might knowledge in this area change my invention?
1		

Write your response here

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Write your response here

| | 2 |

Write your response here

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Write your response here

| | 3 |

Write your response here

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Write your response here

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Think about a user or stakeholder who is very different from you (different age, profession, physical ability, cultural background, economic situation). How might they experience the problem your invention addresses?

Write your response here

What aspect of your invention are you most uncertain about? (This is where your generative model has the weakest predictions.)

Write your response here

Based on this blind spot analysis, what is one thing you should learn more about before proceeding with your invention?

Write your response here

Part 4: Multi-Agent Invention Exercise (15 minutes)

If you have a partner, complete this section together. If not, construct a hypothetical partner based on someone you know from a very different background.

Your Partner's Profile

Partner's name and background:

Write your response here

Partner's top 3 domain knowledge areas:

Write your response here

Partner's likely solution tendencies (what type of solutions would they naturally generate?):

Write your response here

The Collaboration Exercise

Present your invention idea to your partner (real or imagined). Then answer:

What questions does your partner ask that you had not considered?

Write your response here

What alternative solutions does your partner suggest that your generative model would not have generated?

Write your response here

Where do your generative models disagree? (Different assumptions about how the world works, what users want, what is feasible)

Write your response here

Where do your generative models align? (Shared understanding of the problem, common goals, agreed-upon constraints)

Write your response here

What would a merged generative model look like — combining the best of both your perspectives?

Write your response here

Part 5: Your Inventor Agent Profile (10 minutes)

Synthesize the previous parts into a one-page profile.

My Inventor Agent Profile

Identity Statement: I am an inventor who...

Write your response here

Core Generative Model: I understand the world primarily through the lens of...

Write your response here

Top 3 Strengths (what my model predicts well): 1.

Write your response here

1.

Write your response here

1.

Write your response here

Top 3 Blind Spots (what my model misses): 1.

Write your response here

1.

Write your response here

1.

Write your response here

Preferred States (what I am trying to bring about in the world):

Write your response here

Solution Tendencies (the kinds of solutions I naturally gravitate toward):

Write your response here

Ideal Collaborator (what kind of agent would complement my model):

Write your response here

One Belief to Examine (a prior that might be limiting my inventive capacity):

Write your response here

Discussion and Debrief

1. **Generative model diversity**: How did comparing your model with a partner's (or hypothetical partner's) reveal limitations in your own thinking? What specific assumption was most challenged?

2. **Identity and flexibility**: How strongly does your inventor identity constrain your solution space? Is this constraint mostly helpful (providing focus) or limiting (creating blind spots)?

3. **Preferred states and motivation**: How do your multi-scale preferred states (immediate, project, identity) drive your daily inventive decisions? When they conflict, which scale wins?

4. **The active path**: Active Inference distinguishes between changing your expectations (accommodation) and changing the world (active inference). Where in your invention process are you accommodating when you should be acting, or vice versa?

5. **Team composition**: If you were assembling a team for your invention, what generative models (backgrounds, skills, perspectives) would you want represented? Why those specifically?

Write your response here